

~~typed boring notes~~ -> my notes!



RAVINEESH SINGH



Ravineesh_Singh_Resume.pdf

~ 13 Sep 2026 ~

- Howrah, West Bengal, India • +91 7980561223 • ravineeshsingh23@gmail.com
- LinkedIn • GitHub • Portfolio • LeetCode
- Computer Science & **Engineering** Undergraduate at IEST Shibpur with an 8.81/10 CGPA, combining strong analytical problem solving with technical depth in **Data** Structures & Algorithms, Java and C# development, **Enterprise** **Data** Pipelines, and Applied Machine **Learning**. Experienced in designing scalable **data** ingestion workflows, training deep **learning** architectures (CNNs), and applying structured, secure **engineering** practices to deliver high-impact, customer-focused solutions for **enterprise** and financial services environments.

Education *

- Indian Institute of **Engineering** Science and Technology (IEST), Shibpur
- West Bengal, **India**
- Bachelor of Technology in Computer Science and **Engineering** CGPA: 8.81 / 10
- Aug 2024 - May 2028
- Higher Secondary **Education** (Class XII) - UPMSP
- **Uttar Pradesh**, **India**
- **Uttar Pradesh** Board of High School and **Intermediate** **Education** Percentage: 83.6%
- Apr 2021 - May 2023
- Secondary **Education** (Class X) - UPMSP
- **Uttar Pradesh**, **India**
- **Uttar Pradesh** Board of High School and **Intermediate** **Education** Percentage: 89.17%
- Apr 2019 - May 2021

Featured Projects *

- Real Estate Price **Predictor** & Market Analytics Platform
- **Python**, Pandas, Scikit-learn, SQL **GitHub**
- AI-Driven Precision **Agriculture** Weed Detection CNN
- **Python**, Keras, **TensorFlow**, **Deep Learning** **GitHub**

Technical & Professional Skills *

- Python, **Java**, C#, C++, C, SQL, JavaScript (Basics)
- Enterprise Application Development (**Java** C#), Cloud **Concepts** AWS / Azure
- **Fundamentals** RESTful APIs, ETL Pipelines, Relational Database Design (DBMS)
- **Convolutional** Neural Networks (CNNs), Deep Learning, Predictive Modeling, Regression,
- **Classification** Scikit-learn, Keras, TensorFlow
- Pandas, NumPy, Matplotlib, Seaborn, Business Intelligence (Power BI / Tableau **concepts**),
- Exploratory **Data** Analysis (EDA)
- **Data** Structures & Algorithms (DSA), Object-Oriented Programming (OOP with **Java**/ C#),
- Git, GitHub, VS Code, Linux/Bash, Command Line / Terminal (CLI), Agile / Scrum, SDLC

Certifications & Coursework *

Leadership & Community Engagement *

- Core Member - Icche (Social Welfare Club, IEST Shibpur)
- IEST Shibpur
- ETL & **Data** Ingestion Pipeline: Engineered an automated web scraping and ETL pipeline in Python to extract, clean, and normalize 1,000+ multi-attribute property records from 99acres.com, ensuring schema integrity.
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- Feature **Engineering** & Machine Learning: Architected end-to-end **data** preprocessing workflows incorporating outlier detection, categorical encoding, feature scaling, and multi-algorithm regression (Linear, Ridge, Random Forest) to forecast valuations with high precision.
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- Business Intelligence & Market Insights: Developed exploratory **dataanalysis** (EDA) frameworks to model price elasticity, geographical valuation trends, and commercial demand indicators for data-backed decision making.
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- Dataset **Engineering** & Augmentation: Curated and annotated a domain-specific dataset of 2,000 high-resolution agricultural images, applying image transformation and augmentation pipelines (zoom, shear, rotation) to prevent overfitting.
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- Convolutional Neural Network (CNN): Designed and trained a custom deep CNN architecture in Keras utilizing Batch Normalization and Dropout regularization ($p=0.4$) for high-accuracy feature extraction and real-time classification.

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- Model Optimization & Edge Deployment: Benchmarked model performance using Precision, Recall, F1-Score, and Confusion Matrix, targeting edge deployment to automate targeted herbicide spraying and drive sustainable agribusiness efficiencies.
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- **Programming** languages:
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- Cloud & Modern Enterprise:
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- **Data** Science & AI / ML:
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- **Data** Analysis & Visualization:
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- Software **Engineering** Tools:
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- Command-Line & Terminal Fundamentals: Completed a hands-on Terminal course covering shell navigation, file and directory operations, and core command-line utilities for efficient development workflows (Caddy).
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- **Data** Structures & Algorithms (DSA): Comprehensive coursework in algorithmic problem solving, computational complexity optimization, dynamic **programming** and tree/graph **structures** (Udemy).
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- Core Academic Coursework: **Data** Structures & Algorithms, Design & **Analysis** of Algorithms, Object-Oriented **Programming**, Database Management Systems (DBMS), Operating Systems, Discrete Mathematics.
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- Educational Outreach & Mentorship: Volunteer 3 days weekly to teach foundational STEM and literacy to underprivileged K-12 students, fostering educational equity and measurable academic improvement.
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- Project & Event Management: Lead cross-functional student volunteer teams to plan and execute campus-wide community welfare drives, logistics for annual Sports Days, and regional clothing/relief campaigns.
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